

MASI Fundamental Value Estimation and Agent-Based Modeling

PFE Finance Project

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1 Project Overview

This project focuses on the estimation of the fundamental value of the Moroccan All Shares Index (MASI) through a combination of econometrics, signal processing, and agent-based modeling (ABM). The goal is to provide a quantitative framework for analyzing market trends and understanding the behavior of heterogeneous agents in the Moroccan stock market.

2 Technical Implementation

The project's codebase is designed for modularity and scalability:

2.1 Data Acquisition (`db_client.py`)

Historical MASI prices are retrieved from a Supabase backend. The ingestion pipeline includes:

- Secure connection management using environment variables.
- Automated pagination and date-based filtering.
- Data validation and cleaning, ensuring a standardized Pandas DataFrame for downstream analysis.

2.2 Fundamental Value Estimation (`signal_processing.py`)

To proxy the unobserved fundamental price (p_t^f), the project employs the Hodrick-Prescott (HP) filter. This allows for the decomposition of observed prices into:

- **Trend Component:** Interpreted as the fundamental value, smoothed to reflect long-term market drivers.
- **Cyclical Component:** Capturing short-term deviations and noise.

The implementation defaults to a smoothing parameter $\lambda = 129,600$ for daily observations, consistent with standard econometric practices.

3 Theoretical Framework

The project is grounded in the literature on Agent-Based Computational Economics (ACE). It specifically references the Chen et al. 2-type framework, which models market dynamics through the interaction of:

- **Fundamentalists:** Agents who trade based on the perceived gap between the market price and the fundamental value.
- **Chartists (Trend Followers):** Agents who extrapolate past price movements.

Relevant research papers included in the project bibliography cover topics such as evaluating fundamental values in agent-based models and the integration of econometrics with ABM.